Coherence-Guided Dead-Head Identification in Frozen Transformers: A Zero-Parameter Geometric Threshold from Coupled-Oscillator Criticality

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We test a narrow hypothesis: dead attention heads in a frozen transformer obey a coherence boundary that can be derived from coupled-oscillator criticality rather than fitted from model-specific calibration. The claim of this paper is an identification law, not a complete removal procedure. The operational observable is head-level coupling between a head's incoming attention-weighted value write-back and the receiver's pre-head residual stream. The geometric interpretation is explicit: LayerNorm places token states approximately on S^{d-1}, attention heads act as state-dependent inter-oscillator couplings, and MLP updates act as intra-oscillator modes. Starting from the lattice Coherence Learning Rule (CLR) bond-death threshold $\cos(\Delta\theta) = 0.679$ on S^1, normalizing by the natural fluctuation scale on the circle, and then rescaling by concentration of measure on S^{d-1} yields a transformer death threshold

$$\boxed{\tau_{\text{death}}(d) = \frac{\chi_c}{\sqrt{d}}}, \qquad \chi_c = 0.679\sqrt{2} = 0.96025\ldots \tag{1}$$

The rounded shorthand used in the prose is therefore $\tau_{\text{death}}(d) \approx 0.96/\sqrt{d}$. No pruning parameter is fitted.

We evaluate the resulting two-pass system on five frozen decoder checkpoints: GPT-2, GPT-2 Medium, Qwen2.5 0.5B, SmolLM2 360M, and OpenLLaMA 7B. The dead-head pass achieves 95.2%, 98.2%, 95.5%, 99.6%, and 100.0% precision, respectively. The threshold therefore transfers from $d = 768$ through $d = 4096$ without recalibration when validated against individual head ablation ground truth. We explicitly do not claim here that simultaneously removing all detected dead heads is lossless, that naive deletion is the correct downstream pruning operator, or that a

universal removal schedule is solved. Secondary structural compaction analyses are included as systems-side context only.

The result is not theorem closure for transformer dynamics. It is a high-precision, zero-parameter dead-head detector derived from the same critical-ratio logic used elsewhere in the coherence-lattice program.

I. INTRODUCTION

Most transformer pruning methods are calibrated empirically. They rank weights, heads, or channels using magnitude, gradient, activation, or one-shot saliency signals, then fit model-specific cutoffs or sparsity schedules [1–7]. That approach can work well, but the threshold itself is not derived. It must be searched, tuned, or inferred again for each architecture.

This paper tests a different question. Suppose a frozen transformer is treated as a dynamical system propagating activity through a fixed coupling field. Is there a derived coherence boundary below which a structured unit is not participating in coherent information flow at all? If such a boundary exists, one obtains a dead-head identification law; downstream pruning then becomes a separate engineering problem rather than another fitted ranking heuristic.

The present result is deliberately narrow.

1. We study *frozen-model dead-head identification*, not training-time sparsity or a full pruning pipeline.

2. We focus on *attention heads* as the first identification axis.

3. We evaluate a *derived death threshold* first, and only then report secondary redundancy and removal analyses.

4. We keep the stronger vector-Kuramoto interpretation as a motivating hypothesis, not a closed theorem.

The central claim is operational: a head whose mean coupling to the residual stream falls below

$$\tau_{\text{death}}(d) = \frac{\chi_c}{\sqrt{d}}, \qquad \chi_c \approx 0.96 \tag{2}$$

is in the dead regime and is a high-confidence dead-head identification, subject to boundary and bridge protections. We show that this rule transfers across five decoder checkpoints under individual head ablation validation. We do not claim in this paper that the full cumulative removal problem is already solved or that the correct downstream operator is naive head excision.

The derivation chain used throughout the paper is explicit:

$$\cos(\Delta\theta)_{\text{death}} = 0.679 \xrightarrow{\div\,(1/\sqrt{2})} \chi_c = 0.96025\ldots \approx 0.96 \xrightarrow{\times\,(1/\sqrt{d})} \tau_{\text{death}}(d) = \frac{\chi_c}{\sqrt{d}}. \tag{3}$$

The first arrow converts the scalar lattice death point at the BKT critical radius into a dimensionless critical ratio on S^1; the second transfers that ratio into the random-alignment scale of S^{d-1}.

II. TRANSFORMER AS A COUPLED OSCILLATOR NETWORK

A. Sphere Geometry from LayerNorm

Let $x_i^{(l)} \in \mathbb{R}^d$ denote the residual-stream state at token position i before transformer layer l. Ignoring the learned affine parameters, LayerNorm acts as a radial reprojection onto an approximately fixed-radius sphere. The normalized token state therefore lives approximately on

$$\hat{x}_i^{(l)} \in S^{d-1}. \tag{4}$$

At block level, the transformer update takes the form

$$\hat{x}_i^{(l+1)} = \mathrm{LN}\Big(\hat{x}_i^{(l)} + \Delta_{\text{attn},i}^{(l)} + \Delta_{\text{mlp},i}^{(l)}\Big), \tag{5}$$

which is a discrete-time *perturb-and-reproject* dynamics on a high-dimensional sphere.

B. Attention Heads as Inter-Oscillator Couplings

For attention head h, the incoming write-back at receiver position i is

$$s_h(i) = \sum_j A_{ij}^{(h)} V_h(x_j). \tag{6}$$

This is the transformer analog of a state-dependent coupling between oscillator states. The comparison point is the vector Kuramoto / Lohe flow on S^{d-1},

$$\frac{dx_i}{dt} = \sum_j K_{ij}\Big[x_j - (x_i \cdot x_j)x_i\Big], \tag{7}$$

where the tangent-space projection keeps the dynamics on the sphere. In the transformer, the attention weights $A_{ij}^{(h)}$ play the role of driven, state-dependent couplings, while LayerNorm provides the discrete reprojection back onto the representation sphere.

C. MLP Channels as Intra-Oscillator Modes

The MLP update does not couple different token positions. Instead it transforms the state of a single token,

$$\Delta_{\mathrm{mlp},i}^{(l)} = W_{\mathrm{down}}\, \phi\Big(W_{\mathrm{up}} x_i^{(l)}\Big). \tag{8}$$

In the oscillator picture, attention heads are therefore *inter-oscillator* transport channels, while MLP channels are *intra-oscillator* modes or local tangent directions of the update field. The present paper concerns only the attention-side coupling channels.

D. Why the Cosine Observable Is the Right One

On S^1, the pairwise Kuramoto alignment observable is $\cos(\theta_j - \theta_i)$. On S^{d-1}, the natural generalization is directional cosine between the incoming signal and the receiver state. The head-level observable used in this paper,

$$c_h(i) = \cos\Big(s_h(i), x_i\Big), \tag{9}$$

therefore measures the same physical quantity in the higher-dimensional setting: how much of the incoming coupling signal is coherent with the receiver's existing state. Below threshold, the coupling is no longer distinguishable from the random alignment scale of S^{d-1}.

III. FROM LATTICE BOND DEATH TO TRANSFORMER HEAD DEATH

A. Lattice Death Threshold

The underlying physical starting point is the Coherence Learning Rule (CLR) on coupled oscillators. For a lattice bond (ij) with coupling K_{ij} and phase difference $\Delta\theta_{ij}$, the CLR

TABLE I. Paradigm mapping between the coherence-lattice / Kuramoto picture and the transformer picture used in this paper.

Lattice / Kuramoto object	Transformer object
Oscillator state on S^1 or S^{d-1}	token-position residual state
Phase / state variable	residual-stream vector at one token position
Bond coupling K_{ij}	attention weight / effective head coupling
Pairwise alignment $\cos(\Delta\theta)$	$\cos(s_h(i), x_i)$
Internal frame / local mode	MLP channel update direction
Bond death below criticality	dead head below $\tau_{\text{death}}(d)$
Phase-locked mode cluster	coherent cross-layer head cluster / circuit
Domain wall / bridge bond	boundary or bridge-critical head

update has the form

$$\dot{K}_{ij} = \eta \left[R_0(K_{ij}) \cos(\Delta\theta_{ij}) - \frac{2K_{ij}}{r} \right], \tag{10}$$

where $R_0(K) = I_1(K)/I_0(K)$ is the modified Bessel ratio. At the alive/dead boundary, the update changes sign. Evaluated at the critical radius of the lattice analysis, this gives the scalar death condition

$$\cos(\Delta\theta)_{\text{death}} = \frac{4}{r} = 0.679. \tag{11}$$

This threshold lives on S^1 and is tied to the same critical coherence logic used in the lattice-side program.

B. Dimensional Transfer by Concentration of Measure

The raw threshold 0.679 does not transfer directly to hidden-state geometry. What transfers is the critical ratio relative to the natural fluctuation scale of random directions.

On S^1, the natural cosine scale is

$$\sigma_1 = \frac{1}{\sqrt{2}}. \tag{12}$$

This defines a dimensionless critical ratio

$$\chi_c = \frac{0.679}{1/\sqrt{2}} \approx 0.96. \tag{13}$$

Numerically this gives $\chi_c = 0.96025\ldots$; throughout the paper we cite $\chi_c \approx 0.96$ as the rounded normalized critical ratio. The frozen publication bundle uses the exact threshold value stored in each JSON artifact, so the bundled normalized thresholds span $0.96000 \leq \sqrt{d}\,\tau_{\text{death}}(d) \leq 0.96025$.

For unit vectors on S^{d-1}, concentration of measure implies

$$x \cdot y \sim \mathcal{N}\Big(0, \frac{1}{d}\Big) \tag{14}$$

to leading order in the random-direction baseline, so the natural fluctuation scale becomes [8]

$$\sigma_d = \frac{1}{\sqrt{d}}. \tag{15}$$

Replacing the scalar fluctuation scale with the d-dimensional one yields the transformer death threshold

$$\boxed{\tau_{\text{death}}(d) = \chi_c \sigma_d = \frac{\chi_c}{\sqrt{d}}.} \tag{16}$$

Equivalently, the full bridge from the scalar BKT-side critical point to the transformer threshold is

$$0.679 \xrightarrow{\div\,(1/\sqrt{2})} 0.96025\ldots \xrightarrow{\times\,(1/\sqrt{d})} \frac{\chi_c}{\sqrt{d}}. \tag{17}$$

Proposition III.1. *If transformer head coupling is measured as a directional alignment on S^{d-1}, then the leading-order transfer of the CLR death boundary from scalar phase space to hidden-state geometry is $\tau_{\text{death}}(d) = \chi_c/\sqrt{d}$ with $\chi_c \approx 0.96$.*

Remark III.1. *This paper validates the transfer empirically across $d = 768$, 896, 960, 1024, and 4096. It does not claim a complete generalized BKT theorem on S^{d-1}.*

C. Transformer Observable

Let x_i be the pre-head residual state at destination position i, and let head h write an incoming signal

$$s_h(i) = \sum_j A_{ij}^{(h)} V_h(x_j), \tag{18}$$

where $A_{ij}^{(h)}$ is the attention weight and V_h is the value pathway of head h. We define the per-position coupling

$$c_h(i) = \cos\Big(s_h(i), x_i\Big) \tag{19}$$

and the head-level mean coupling

$$\bar{c}_h = \mathbb{E}_{\text{batch},\, i}\, c_h(i). \tag{20}$$

The dead-head identification criterion is then

$$\bar{c}_h < \tau_{\text{death}}(d), \tag{21}$$

subject to two protections:

1. **Boundary protection**: the first one or two layers are treated as input transducers, and the final layer may be treated as an output boundary.

2. **Bridge veto**: heads with unusually high bridge mass in the layerwise attention graph are protected as rare connectivity-critical channels.

IV. OPERATIONAL METHOD

The method used in the experiments is intentionally simple.

1. Compute the incoming write-back signal for each head at each receiving position.

2. Measure cosine alignment between that signal and the pre-head residual.

3. Average over positions and calibration sequences to obtain one scalar $\bar{c}_h$ per head.

4. Identify the head as dead if $\bar{c}_h < \tau_{\text{death}}(d)$ and the head is not boundary-protected or bridge-protected.

5. Validate the resulting dead-head set against individual head ablation ground truth. Structural removal experiments are treated as a separate secondary analysis.

This completes the scientific claim tested in the paper. Identification is kept separate from removal: a head can be individually ablation-safe yet still participate in distributed basis rotation across layers when many heads are removed together. Rotation-aware compensation operators, including SVD-based variants, belong to downstream pruning engineering and are not part of the geometric law validated here.

For standard multi-head attention (MHA), structural removal means deleting the relevant $q/k/v/o$ slices for the dead head. For grouped-query attention (GQA), the system distinguishes between:

- **dead query-head removal**, and
- **Level 2 KV-group compaction**, where a full KV group is deleted only when all query heads attached to that KV group are dead.

The current paper uses a second pass only after the dead set is established. Among the heads kept alive by the death rule, we mark heads redundant when their output geometry is contained within other alive heads in the same layer. This redundancy pass is geometry-aware but is not itself claimed as a theorem-derived threshold.

V. EXPERIMENTAL PROTOCOL

A. Models and Calibration

We evaluate five frozen decoder checkpoints:

1. GPT-2 ($d = 768$)
2. GPT-2 Medium ($d = 1024$)
3. Qwen2.5 0.5B ($d = 896$)
4. SmolLM2 360M ($d = 960$)
5. OpenLLaMA 7B ($d = 4096$)

All runs use fixed-length calibration and evaluation slices with small held-out batches for rapid structural ablation. Ground-truth safety is defined from per-head ablation damage using a conservative small-loss threshold proportional to baseline loss. The emphasis of the paper is therefore precision: if the dead-pass marks a head as dead, how often is that head empirically safe to remove?

B. What Is Actually Claimed

The validated claim surface for this manuscript is:

- the observable above is a useful transformer-native dead-head identification observable,

- the threshold $\tau_{\text{death}}(d) = \chi_c/\sqrt{d}$ with $\chi_c \approx 0.96$ transfers across five bundled decoder checkpoints without fitted parameters,
- the primary validation target is individual head ablation precision,
- boundary protection is necessary,
- the paper establishes an identification law, not a universal removal operator,
- secondary systems analyses can produce real speed and KV-cache savings once the runtime path is actually compacted.

The following are not claimed here:

- theorem closure that transformers exactly obey vector Kuramoto dynamics,
- the full generalized BKT critical theory on S^{d-1},
- a universal lossless removal protocol for the identified dead heads,
- that naive deletion or any single compensation operator is already the correct downstream pruning map,
- CLR-during-training as an already-validated training framework.

VI. RESULTS

A. Five-Model Transfer

Table II summarizes the full five-model result surface. The scientific center of gravity is the dead-pass precision column. The redundancy and cumulative removal columns are included as secondary engineering context, not as the primary publication claim.

The main pattern is stable across all five pilots:

- the dead-pass remains high precision,
- the redundancy pass is useful but more architecture-sensitive,
- the threshold continues to scale correctly as d increases.

TABLE II. Five-model transfer of the derived death threshold. Dead precision refers to the first-pass dead-head identification criterion. Redundancy precision refers to the second-pass redundancy detector within the alive pool. Heads-removed and cumulative-loss columns are secondary engineering context under specific removal operators.

Model	Family	d	$\tau_{\text{death}}(d)$	Dead Prec.	Red. Prec.	Heads Removed	Combined Prec.	Dead+Red ΔLoss
GPT-2	GPT-2	768	0.03465	95.2%	88.5%	68 / 144	92.7%	+1.84669
GPT-2 Medium	GPT-2	1024	0.03000	98.2%	100.0%	199 / 384	99.0%	+1.43029
Qwen2.5 0.5B	Qwen2	896	0.03207	95.5%	95.2%	219 / 336	95.4%	+2.12476
SmolLM2 360M	Llama	960	0.03098	99.6%	89.5%	299 / 480	98.3%	+2.05133
OpenLLaMA 7B	Llama	4096	0.01500	100.0%	98.6%	355 / 1024	99.7%	+1.60351

B. Head-Level Threshold Evidence

The threshold behavior is visible directly at head level once coupling is plotted in normalized coordinates

$$z_h = \sqrt{d}\,\bar{c}_h, \tag{22}$$

so that the bundled threshold boundary collapses to the narrow range $0.96000 \leq \chi_c \leq 0.96025$.

Figure 2 shows representative small-scale and LLaMA-scale views. On GPT-2, 58/144 heads fall below the threshold; 41 of those are killed and 13 are retained only because boundary or bridge protection vetoes the death decision. On OpenLLaMA 7B, 338/1024 heads fall below threshold; 273 of those are killed and 34 are protected. The GPT-2 dead-head false positives remain the same two interior layer-5 heads. OpenLLaMA has no dead-head false positives on the current pilot slice.

The normalized separation is also stable. For the five validated models, the dead heads have normalized-coupling medians at or below zero, while the retained alive heads have medians above 1.86 and the redundant heads have medians above 2.72. The decision maps show the same structural pattern seen in the scalar statistics: boundary protection is concentrated in the earliest layers, while dead heads populate the interior of the network.

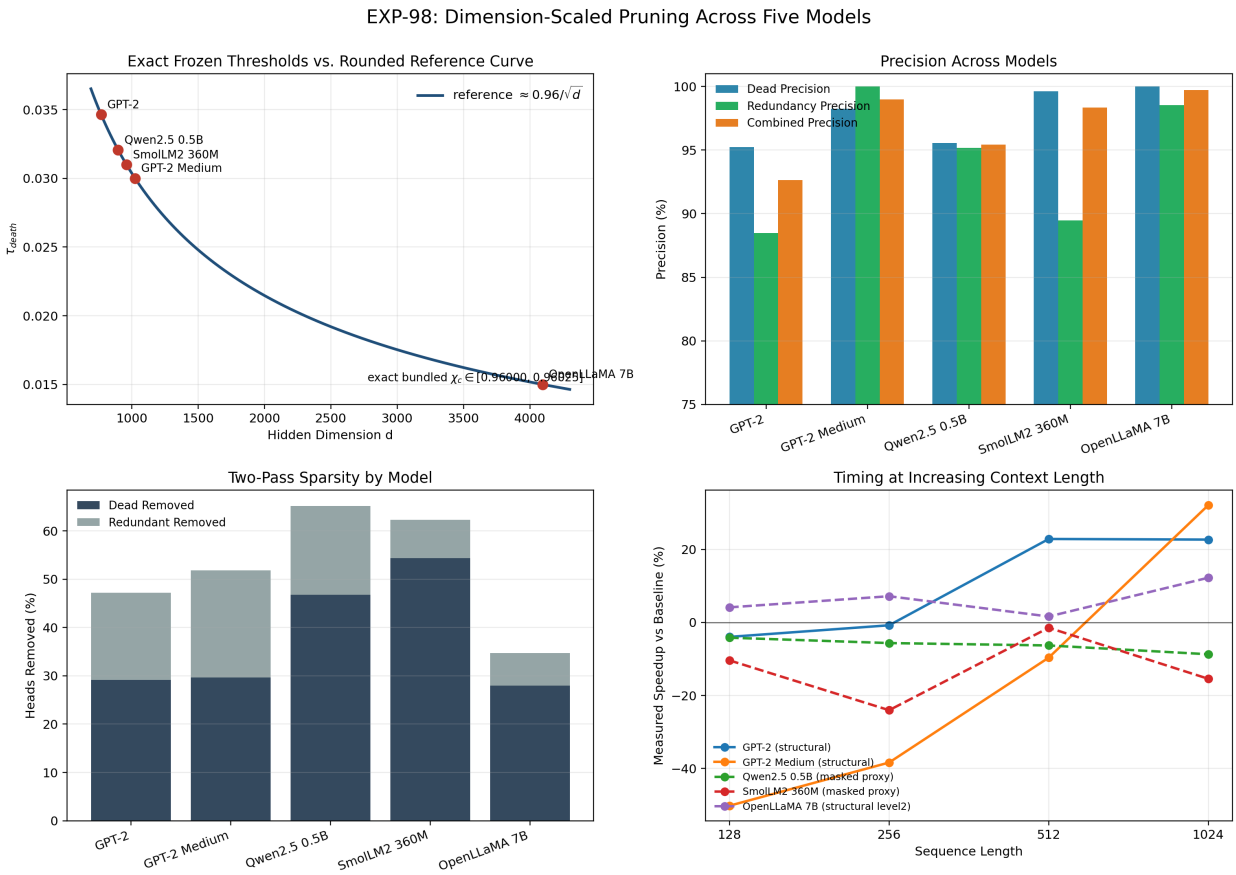

FIG. 1. Five-model summary of the derived threshold curve, dead/redundancy precision, resulting sparsity, and measured timing where structural compaction exists.

C. Normalized Coupling Collapse

Figure 3 shows the same threshold in cross-model normalized coordinates. Once $\bar{c}_h$ is rescaled by $\sqrt{d}$, the raw dimension dependence is removed and the exact bundled thresholds collapse into a narrow band near $z_h \approx 0.96$.

The left panel shows that the five bundled checkpoints occupy comparable normalized ranges despite spanning hidden dimensions from 768 through 4096. The right panel shows what happens to below-threshold heads after the protection rules are applied: the majority become dead-head decisions, while most of the surviving minority is explained by protection rather than by ordinary interior heads remaining alive below threshold.

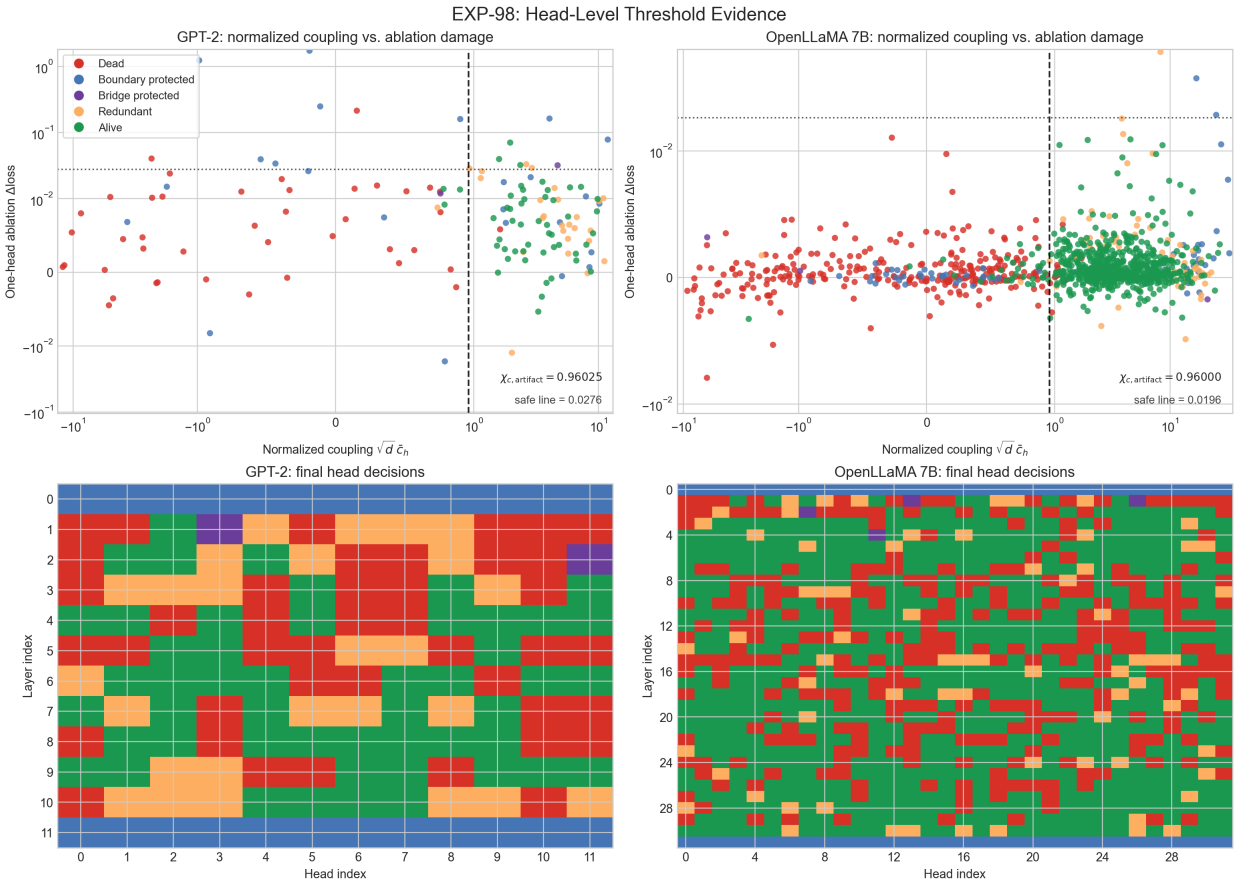

FIG. 2. Head-level threshold evidence. Top row: normalized coupling $z_h = \sqrt{d}\,\bar{c}_h$ versus one-head ablation damage for GPT-2 and OpenLLaMA 7B, with the exact artifact-specific normalized threshold drawn as a vertical dashed line. Bottom row: categorical layer/head decision maps showing dead, protected, redundant, and retained alive heads.

D. Grouped-Query Level 2 Compaction

SmolLM2 is the strongest grouped-query Level 2 target because its group size is only $g = 3$. The dead-head map induces:

$$44/160 = 27.5\% \tag{23}$$

fully dead KV groups, with

$$132/480 = 27.5\% \tag{24}$$

associated query heads removed. This yields:

- parameter count reduction: 361.8M $\rightarrow$ 340.2M (-5.98%),

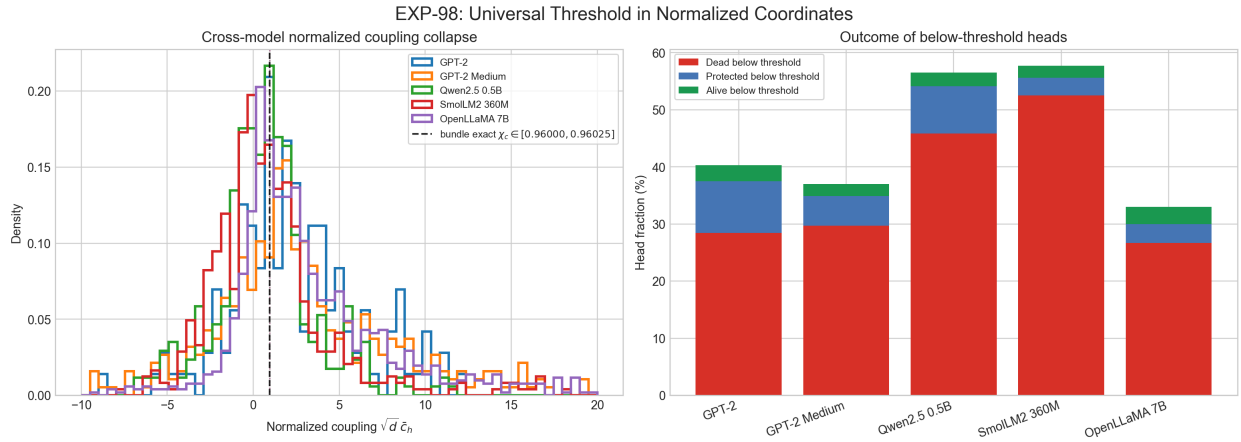

FIG. 3. Cross-model normalized coupling collapse. Left: all-head density in normalized coordinates for the five completed models with the exact bundled thresholds clustered at $\chi_c \in [0.96000, 0.96025]$. Right: model-wise breakdown of below-threshold heads into dead, protected, and still-alive outcomes.

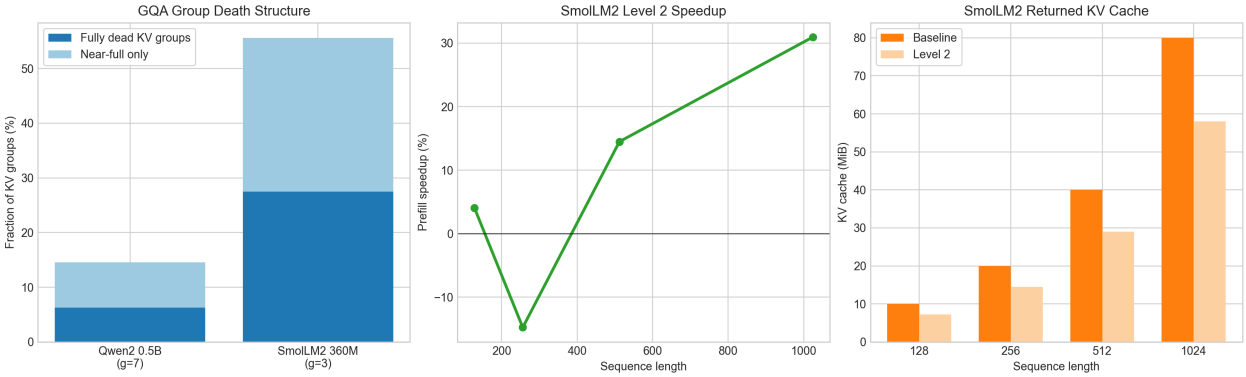

FIG. 4. Focused grouped-query result. SmolLM2's smaller group size makes full KV-group death common enough for true Level 2 structural compaction; Qwen2's larger group size does not.

- returned KV-cache reduction per token: 81,920 → 59,392 (−27.5%),
- measured prefill speedup at sequence length 1024: +30.96%,
- Δloss at sequence length 1024: +0.53206.

The Qwen2 comparison is informative. With group size $g = 7$, only 3/48 KV groups are fully dead. The group-size barrier suppresses pure Level 2 compaction combinatorially. This is a structural reason to separate the threshold story from the systems story.

TABLE III. Timing summary at sequence length 1024.

Model	Timing Mode	Variant	Speedup vs. Baseline	ΔLoss
GPT-2	structural	dead+redundant	+22.69%	+2.66609
GPT-2 Medium	structural	dead+redundant	+32.14%	+2.19939
Qwen2.5 0.5B	masked proxy	dead+redundant	-8.72%	+2.49661
SmolLM2 360M	structural Level 2	fully dead KV groups	+30.96%	+0.53206
OpenLLaMA 7B	structural dead-only	dead heads	+12.25%	+1.15692

E. LLaMA-Scale Validation

The OpenLLaMA 7B row is the key scaling test. At $d = 4096$,

$$\tau_{\text{death}}(4096) = \frac{\chi_c}{\sqrt{4096}} \approx 0.015, \tag{25}$$

and the dead pass produces:

- 286/1024 dead heads,
- 100.0% dead-pass precision,
- zero dead-head false positives on the current pilot slice.

Because OpenLLaMA 7B uses full multi-head attention ($g = 1$), dead-head structural compaction is already the Level 2 result. The measured structural outcome is:

- parameter reduction: 6.74B → 6.14B (−8.90%),
- KV cache reduction: 524,288 → 377,856 bytes per token (−27.93%),
- measured speedup at sequence length 1024: +12.25%,
- Δloss at sequence length 1024: +1.15692.

F. Timing Matrix

Table III summarizes the strongest sequence-length-1024 timing rows.

This cleanly separates the classification result from the runtime result:

- the formula already transfers across architectures,
- runtime gains appear where true structural compaction exists,
- large-group GQA still needs Level 3 rebalancing to unlock the same systems payoff.

VII. DISCUSSION

The strongest result in this paper is not that coherence "wins a ranking horse race." It is that a threshold derived from oscillator criticality and high-dimensional geometry identifies dead heads at high precision across five architecturally distinct frozen decoders.

The dead-pass behavior is conservative by design. It finds the definitely dead subset first. Many safe-but-kept heads are coherent yet redundant. That is why the redundancy pass is treated as a distinct compaction mechanism rather than as a fix to the death threshold itself.

The cumulative removal rows should therefore not be read as a lossless-pruning claim. They are secondary measurements of what happens under particular removal choices once the dead set has been identified.

The paper also clarifies a systems point. Transfer of the threshold and transfer of runtime speed are not the same event. Qwen2 validates the classifier but still lacks native grouped-query structural compaction. SmolLM2 validates both the classifier and the first grouped-query structural pathway. OpenLLaMA 7B validates the large-d transfer and a true structural speedup at LLaMA scale.

The same separation applies to cumulative removal quality. A head can satisfy the dead-head identification law and still require compensation when many heads are removed together, because multi-head deletion can perturb distributed basis rotations across layers. That is why this paper validates the law with one-head ablation precision rather than treating naive cumulative deletion as the theorem target. Rotation-aware pruning operators, including SVD-based compensation schemes, belong to follow-on engineering work.

Separate later CLR pipeline runs also continue to surface sizeable dead-head sets at larger scale, e.g. 407 dead heads (26.5%) on REAP-25B MoE. We do not elevate those rows into the main claim here because this publication bundle does not include their matching per-head ablation ground truth.

Two natural follow-on papers are now separated from the present result:

1. a generalized BKT / vector-Kuramoto critical theory on S^{d-1},
2. CLR-during-training, where dead heads and layers are removed online rather than post hoc.

Both are legitimate next steps, but neither is needed to support the present publication claim.

A. Independent Confirmation from Weight Compression

An independent line of evidence strengthens the geometric picture. Turney [9] recently conducted systematic A/B isolation of tensor roles in transformer weight compression, discovering through brute-force ablation that:

1. attention tensors tolerate WHT-rotated quantization well,
2. FFN write-back projections (`ffn_down`) are quality-critical, and
3. boundary layers are disproportionately sensitive.

These three findings were reached without reference to the CLR framework. Each maps directly onto elements of the present paper:

- attention tolerance $\leftrightarrow$ inter-oscillator couplings operate near the random-alignment scale, so moderate perturbation does not shift them across the death boundary;
- write-back sensitivity $\leftrightarrow$ the residual-stream output projection is the coupling's *write* channel into the shared state, where errors compound layer-over-layer (same mechanism as K-error amplification through softmax in KV cache compression);
- boundary sensitivity $\leftrightarrow$ boundary layers act as driven lattice boundary nodes under different dynamics from interior layers.

We applied the tensor-role insight to the CLR weight-quantization stage of our compression pipeline (Stage 4b). Guided by the anatomy eligibility profile from the *same single forward pass* used for dead-head identification, attention and FFN read projections receive

4-bit WHT-rotated quantization while FFN write-back projections receive 6-bit protection. On Qwen3-8B ($d = 4096$, 36 layers), this role-aware policy reduces the weight-quantization quality cost from $\Delta\text{Loss} = +0.339$ (uniform 4-bit) to $\Delta\text{Loss} = +0.038$ — an **89% reduction in quality loss** at comparable effective bit rate (4.24 vs 3.69 bits). No additional calibration, retraining, or Hessian computation is required; the anatomy profile already contains the information needed.

This result is relevant to the present paper not as a pruning claim but as *convergent evidence*: the same coupled-oscillator geometry that identifies dead heads also predicts which weight matrices tolerate compression and which require protection. The tensor-role asymmetry discovered empirically by Turney is a natural consequence of the distinction between inter-oscillator transport channels (attention) and intra-oscillator modes (MLP) articulated in Section II C.

VIII. CONCLUSION

We presented a zero-parameter dead-head identification criterion for frozen transformers:

$$\boxed{\tau_{\text{death}}(d) = \frac{\chi_c}{\sqrt{d}}}, \qquad \chi_c \approx 0.96. \tag{26}$$

The threshold is derived, not fitted. Its observable is the mean cosine alignment between a head's incoming write-back signal and the receiver's pre-head residual state. Across GPT-2, GPT-2 Medium, Qwen2.5 0.5B, SmolLM2 360M, and OpenLLaMA 7B, the dead-pass reaches 95.2% through 100.0% precision with no model-specific recalibration when checked against individual head ablation. Structural compaction results are secondary analyses and are not the core scientific claim of the paper.

The current paper therefore establishes a clean result: the coherence death boundary transfers from lattice criticality into transformer head-death identification through dimensional scaling. That is a meaningful publication result even before the wider removal, generalized BKT, and training-time programs are developed.

Appendix A: Reproducibility Package

This publication folder is structured as a local package.

- `paper.tex`: formal manuscript source
- `references.bib`: bibliography
- `data/`: frozen JSON result artifacts
- `figures/`: bundled figures
- `scripts/98_result_plots.py`: regenerate the five-model summary
- `scripts/98_gqa_level2_plot.py`: regenerate the grouped-query figure
- `scripts/98_threshold_evidence_plots.py`: regenerate the head-level threshold evidence and normalized-collapse figures
- `scripts/98_coherence_pruning_harness.py`: rerun the head-scan and theory classification pass
- `scripts/98_verify_threshold_bundle.py`: verify the five bundled transfer rows and write the machine-readable threshold summary
- `scripts/98_structural_timing_benchmark.py`: rerun structural timing
- `scripts/98_smollm2_kv_group_compaction.py`: rerun grouped-query Level 2 compaction

The included JSON artifacts are sufficient to rebuild the publication tables and figures without rerunning the models. Full model reruns require access to the relevant Hugging Face checkpoints and a calibration/evaluation corpus.

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